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SEAT No. _____

No. of Printed Pages: 02

Sardar Patel University
External Examination
B. Sc. (CA & IT) (Integrated) M. Sc. IT (Integrated) – 1st Semester
PS01CIIIT03: Problem Solving Methodology and Programming in C
29th October, Monday – 2018

Time: 10:00 am to 1:00 pm

Total Marks: 70

Q.1 Select an appropriate answer.

10

1. _____ is a step by step approach to solve any problem.
(a) Process (b) Programming Language (c) Algorithm (d) Compiler
2. _____ is a pictorial representation of an algorithm.
(a) Data Diagram (b) Flow Chat (c) Pie Chart (d) Program
3. What is the standard decision symbol for a flowchart?
(a) circle (b) lozenge (c) diamond (d) square
4. The range of int data type is _____ bytes.
(a) 2 (b) 4 (c) 8 (d) 16
5. The value can be changed during program execution is known as _____.
(a) Variable (b) constant (c) Operator (d) None of these
6. _____ format specifier used for integer data type.
(a) %d (b) %f (c) %c (d) %e
7. In array subscript can begin with number _____.
(a) Zero (b) One (c) three (d) None of these
8. '\0' is _____.
(a) Null character (b) Character value
(c) Escape sequence (d) Symbolic constant
9. A group of character is known as _____.
(a) String (b) Array (c) Function (d) All of above
10. The _____ string function is use to joins two string.
(a) strcat() (b) strlen() (c)strupr() (d) strrrev()

Q.2 Answer in short. (Attempt any TEN)

20

1. What is Translator? List all translators.
2. What is an Editor? Give 3 examples of Well Known Editors.
3. Write an algorithm to find simple interest.
4. Write rules of variable name.
5. Explain ternary operator with example.
6. Write difference between pre-increment and post- increment operator.

(1)

[P.T.O.]

7. Explain break and continue statement in C.
8. Explain do-while statement with example.
9. Explain structure programming.
10. What is string? List out the operation perform on the string.
11. What is user define function? Give one example of user define function.
12. Write the advantage of function.

Q.3

- (a) What is an algorithm? Write advantages and disadvantages of an algorithm. 5
- (b) Write an algorithm to check whether inputted number is prime number or not. 5

OR

Q.3

- (a) What is flow chart? Write advantages and disadvantages of flowchart. 5
- (b) Write an algorithm to check whether inputted number is Armstrong number or not. 5

Q.4

- (a) Explain Basic Structure of C program. 5
- (b) Explain if..else & nested if statement with syntax and example. 5

OR

Q.4

- (a) What is operator? Explain different operators with example. 5
- (b) Explain basic data types used in C language. 5

- Q.5 What is an array? Explain in detail 1D and 2D array with syntax and example. 10

OR

- Q.5 Explain following function with syntax and example: 10
- (i) islower() (ii) pow() (iii) sqrt() (iv) isdigit() (v) abs()

Q.6

- (a) What is function? Explain function definition with syntax and example. 5
- (b) Explain the following category of function with example: 5
 - (i) with argument and no return value
 - (ii) with no argument and return value

OR

Q.6

- (a) Explain formal parameter and actual parameter with example. 5
- (b) Explain command line argument with example. 5

